



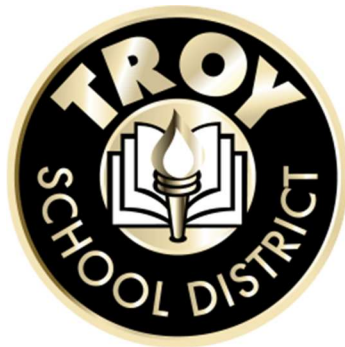
BELL TIME STUDY

SCHOOL BUS LOGISTICS

Date: 4/24/25

Version: 1.0

Prepared for: Troy School District



Thursday, April 24, 2025

Dan Trudel
Troy School District
4400 Livernois
Troy, MI 48098

Dear Mr. Trudel,

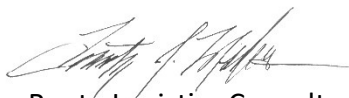
School Bus Logistics, LLC is pleased to present this summary report for the bell time study. This report provides a comprehensive analysis of four distinct scenarios, focusing on the impact of each scenario and the additional vehicle requirements.

I would like to thank you and Kathy for your collaboration throughout this process. The open communication has significantly enhanced our understanding of the district's needs and those of the families it serves.

Should you have any further questions or require additional information regarding this report, please do not hesitate to reach out. I can be contacted at 331-575-2453 or via email at tim.micheals@schoolbuslogistics.com.

Regards,

Timothy Micheals

A handwritten signature in black ink, appearing to read "Timothy Micheals", written in a cursive style.

Route Logistics Consultant

1.0 EXECUTIVE SUMMARY

Study Overview

School Bus Logistics (SBL) has been selected to conduct a bell time study for Troy School District, evaluating four distinct scenarios. At the request of the school board, each scenario will include a later start time for all high school students. This study is grounded in a comprehensive analysis of the current baseline data, along with detailed evaluations of the four proposed scenarios.

Project Objectives:

- Assess the current fleet schedule to establish a baseline
- Develop route packages aligned with bell time options for each scenario
- Ensure each scenario incorporates a later high school start time

For each scenario, SBL will provide:

- Proposed bell times, including AM arrivals and PM departures
- Analysis of vehicle requirements
- Daily driver hours associated with each route package
- Daily vehicle mileage for each route package

Methodology

School Bus Logistics utilized Versatrans RP routing software to develop route sets for each scenario, with a focus on General Education transportation. All data was housed on Troy School District's servers, and SBL accessed it via a secure remote connection to compile and analyze results within Versatrans RP.

To initiate the analysis, SBL established a baseline reflecting the current routing structure and configuration of route packages. Prior to conducting any in-depth analysis, data verification and cleanup were required. This process addressed elements such as anchor sit times, stop sit times, and student load times.

Once finalized, the baseline data was documented using the following criteria:

- Service windows at schools- unloading and loading at schools
- Total number of eligible riders (utilization of seat capacity)
- Actual load counts provided from Troy School District
- Current route pairings

Baseline Data

SBL has performed an analysis based on data from the 2024–2025 school year, which serves as the baseline. The baseline dataset includes key metrics such as vehicle requirements, travel time, and mileage.

Bell Times currently in place for SY 24-25

Tier	School	Arrival AM	Bell Time AM	Bell Time PM	Instructional Min (HH:MM)	Departure PM	# Buses
1	Athens HS	6:48 ~ 7:10	7:20:00 AM	2:08:00 PM	06:48	2:17:00 PM	9
1	Troy HS	6:55 ~ 7:05	7:20:00 AM	2:08:00 PM	06:48	2:17:00 PM	20
2	Larson MS	7:25 ~ 7:50	7:52:00 AM	2:40:00 PM	06:48	2:45:00 PM	9
2	Boulan MS	7:19 ~ 7:45	7:52:00 AM	2:40:00 PM	06:48	2:45:00 PM	14
3	Baker MS	7:50 ~ 8:10	8:17:00 AM	3:05:00 PM	06:48	3:10:00 PM	9
3	Smith MS	7:51 ~ 8:10	8:17:00 AM	3:05:00 PM	06:48	3:10:00 PM	11
4	Bemis ES	8:20:00 AM	8:39:00 AM	3:37:00 PM	06:58	3:44:00 PM	7
4	Hill ES	8:35:00 AM	8:39:00 AM	3:37:00 PM	06:58	3:44:00 PM	1
4	Martell ES	8:24:00 AM	8:39:00 AM	3:37:00 PM	06:58	3:44:00 PM	7AM/8PM
4	Troy Union ES	8:25:00 AM	8:39:00 AM	3:37:00 PM	06:58	3:44:00 PM	7
4	Wattles ES	8:24:00 AM	8:39:00 AM	3:37:00 PM	06:58	3:44:00 PM	7
5	Barnard ES	9:00:00 AM	9:09:00 AM	4:07:00 PM	06:58	4:14:00 PM	4
5	Costello ES	8:55:00 AM	9:09:00 AM	4:07:00 PM	06:58	4:17:00 PM	3
5	Morse ES	9:00:00 AM	9:09:00 AM	4:07:00 PM	06:58	4:14:00 PM	5
5	Hamilton ES	9:03:00 AM	9:09:00 AM	4:07:00 PM	06:58	4:14:00 PM	6AM/5PM
5	Schroeder ES	9:05:00 AM	9:09:00 AM	4:07:00 PM	06:58	4:14:00 PM	7
5	Leonard ES	8:52:00 AM	9:09:00 AM	4:07:00 PM	06:58	4:14:00 PM	5

- 32 vehicles are required for Gen Ed routing utilizing the current 5-tier system
- Daily labor hours of 153
- Daily mileage of 1161

Through the evaluation of current route packages and the district’s transportation needs, four key tasks were identified. Each is summarized in this report.

2.0 TASKS

2.1 Task 1 – Scenario A (High School Time Shift)

SBL initially explored the impact of shifting the current high school arrival time from 7:00 AM to 8:00 AM. Under this scenario, middle school arrival times would adjust accordingly—Boulan Park and Larson would move to a 7:00 AM arrival, while Baker and Smith would shift to 7:30 AM. Elementary school arrival times would remain unchanged. To accommodate this adjustment, fleet schedule pairings were modified, while maintaining the existing routing structure.

Task 1 Summary:

- Bell Times utilized in Scenario A:

Tier	School	Arrival (earliest)	AM Bell Time AM	Bell Time PM	Ins. (HH:MM)	Min Departure PM	# Buses	Gen Ed
1	Larson MS	7:00:00 AM	7:20:00 AM	2:08:00 PM	6:48:00	2:17:00 PM	9	
1	Boulan MS	7:00:00 AM	7:20:00 AM	2:08:00 PM	6:48:00	2:17:00 PM	14	
2	Baker MS	7:30:00 AM	7:52:00 AM	2:40:00 PM	6:48:00	2:47:00 PM	9	
2	Smith MS	7:30:00 AM	7:52:00 AM	2:40:00 PM	6:48:00	2:47:00 PM	11	
3	Athens HS	8:00:00 AM	8:17:00 AM	3:05:00 PM	6:48:00	3:10:00 PM	9	
3	Troy HS	8:00:00 AM	8:17:00 AM	3:05:00 PM	6:48:00	3:10:00 PM	20	
4	Bemis ES	8:20:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7	
4	Hill ES	8:35:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	1	
4	Martell ES	8:24:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7AM/8PM	
4	Troy Union ES	8:25:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7	
4	Wattles ES	8:24:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7	
5	Barnard ES	9:00:00 AM	9:09:00 AM	4:07:00 PM	6:58:00	4:14:00 PM	4	
5	Costello ES	8:55:00 AM	9:09:00 AM	4:07:00 PM	6:58:00	4:14:00 PM	3	
5	Morse ES	9:00:00 AM	9:09:00 AM	4:07:00 PM	6:58:00	4:14:00 PM	5	
5	Hamilton ES	9:03:00 AM	9:09:00 AM	4:07:00 PM	6:58:00	4:14:00 PM	6AM/5PM	
5	Schroeder ES	9:05:00 AM	9:09:00 AM	4:07:00 PM	6:58:00	4:14:00 PM	7	
5	Leonard ES	8:52:00 AM	9:09:00 AM	4:07:00 PM	6:58:00	4:14:00 PM	5	

- 39 vehicles are required for Gen Ed routing utilizing the 5-tier system (7 Additional Vehicles)
- Daily labor hours of 197
- Daily mileage of 1161

2.2 Task 2 – Scenario B (Elementary & Middle School Switch)

Scenario B was developed by SBL using the route set from Scenario A, with the primary adjustment being the reversal of the middle school and elementary school tiers. After a thorough review and discussion between SBL and the Troy School District, it was determined that this scenario was not feasible. The required increase in vehicle count and the potential for creating latchkey situations for elementary students presented significant challenges. While fleet schedule pairings were modified to reflect the proposed tier shift, the existing routing remained unchanged.

Task 2 Summary:

- Bell Times utilized in Scenario B:

Tier	School	Arrival (earliest)	AM Bell Time AM	Bell Time PM	Ins. Min (HH:MM)	Departure PM	# Buses	Gen Ed
1	Barnard ES	7:00:00 AM	7:10:00 AM	2:08:00 PM	6:58:00	2:17:00 PM	4	
1	Costello ES	7:00:00 AM	7:10:00 AM	2:08:00 PM	6:58:00	2:17:00 PM	3	
1	Morse ES	7:00:00 AM	7:10:00 AM	2:08:00 PM	6:58:00	2:17:00 PM	5	
1	Hamilton ES	7:00:00 AM	7:10:00 AM	2:08:00 PM	6:58:00	2:17:00 PM	6AM/5PM	
1	Schroeder ES	7:00:00 AM	7:10:00 AM	2:08:00 PM	6:58:00	2:17:00 PM	7	
1	Leonard ES	7:00:00 AM	7:10:00 AM	2:08:00 PM	6:58:00	2:17:00 PM	5	
2	Bemis ES	7:30:00 AM	7:42:00 AM	2:40:00 PM	6:58:00	2:47:00 PM	7	
2	Hill ES	7:30:00 AM	7:42:00 AM	2:40:00 PM	6:58:00	2:47:00 PM	1	
2	Martell ES	7:30:00 AM	7:42:00 AM	2:40:00 PM	6:58:00	2:47:00 PM	7AM/8PM	
2	Troy Union ES	7:30:00 AM	7:42:00 AM	2:40:00 PM	6:58:00	2:47:00 PM	7	
2	Wattles ES	7:30:00 AM	7:42:00 AM	2:40:00 PM	6:58:00	2:47:00 PM	7	
3	Athens HS	8:00:00 AM	8:17:00 AM	3:05:00 PM	6:48:00	3:10:00 PM	9	
3	Troy HS	8:00:00 AM	8:17:00 AM	3:05:00 PM	6:48:00	3:10:00 PM	20	
4	Larson MS	8:30:00 AM	8:49:00 AM	3:37:00 PM	6:48:00	3:44:00 PM	9	
4	Boulan MS	8:30:00 AM	8:49:00 AM	3:37:00 PM	6:48:00	3:44:00 PM	14	
5	Baker MS	9:00:00 AM	9:19:00 AM	4:07:00 PM	6:48:00	4:14:00 PM	9	
5	Smith MS	9:00:00 AM	9:19:00 AM	4:07:00 PM	6:48:00	4:14:00 PM	11	

- 43 vehicles are required for Gen Ed routing utilizing the 5-tier system (11 Additional Vehicles)
- Daily labor hours of 215
- Daily mileage of 1161

2.3 Task 3 – Scenario C (Middle / High School Combination)

Scenario C was developed by SBL in response to feedback from the Troy School District. This option involved combining middle and high school students on the same bus routes, which increased the eligible rider load per vehicle. Specifically, Larsen and Baker Middle School students were paired with Athens High School students, while Boulan Park and Smith Middle School students were paired with Troy High School students. To support this configuration, new route sets were created, and routing criteria were adjusted to accommodate the combined student populations. New routes were developed to transport students to and from multiple schools on a single bus.

Task 3 Summary:

- Bell Times utilized in Scenario C:

Tier	School	Arrival AM (earliest)	Bell Time AM	Bell Time PM	Ins. Min (HH:MM)	Departure PM	# Gened Buses
1	Larsen MS	7:30:00 AM	7:52:00 AM	2:40:00 PM	6:48:00	2:47:00 PM	Shared / 24
1	Baker MS	7:30:00 AM	7:52:00 AM	2:40:00 PM	6:48:00	2:47:00 PM	Shared / 24
1	Athens HS	8:00:00 AM	8:17:00 AM	3:05:00 PM	6:48:00	3:10:00 PM	Shared / 24
1	Boulan MS	7:30:00 AM	7:52:00 AM	2:40:00 PM	6:48:00	2:47:00 PM	Shared / 37
1	Smith MS	7:30:00 AM	7:52:00 AM	2:40:00 PM	6:48:00	2:47:00 PM	Shared / 37
1	Troy HS	8:00:00 AM	8:17:00 AM	3:05:00 PM	6:48:00	3:10:00 PM	Shared / 37
2	Bemis ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7
2	Hill ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	1
2	Martell ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7AM/8PM
2	Troy Union ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7
2	Wattles ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7
2	Barnard ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	4
2	Costello ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	3
2	Morse ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	5
2	Hamilton ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	6AM/5PM
2	Schroeder ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	7
2	Leonard ES	8:30:00 AM	8:39:00 AM	3:37:00 PM	6:58:00	3:44:00 PM	5

- 56 vehicles are required for Gen Ed routing utilizing the 2-tier hybrid system (24 Additional Vehicles)
- Daily labor hours of 181
- Daily mileage of 1387

An option to implement a 4-tier hybrid system—which would stagger drop-off and pickup times—was briefly explored. This approach had the potential to reduce the number of vehicles needed to an estimated 37–39 buses. However, when this scenario was presented to the school board, a decision was made not to proceed with combining middle and high school students on the same bus, and the option was set aside.

2.4 Task 4 – Scenario D (Node Stops)

Scenario D was developed by SBL as a variation of Scenario A, focused on implementing a node-based stop system utilizing strategically placed neighborhood-based stops for students. This approach aimed to consolidate bus stops and increase walking distances for high school students, thereby reducing the number of individual stops. While the node-based consolidation resulted in a reduction in total mileage, it did not lead to a decrease in driver labor hours. Similar outcomes would be expected if applied to the elementary and middle school tiers. Because this option did not reduce the number of vehicles required, it was not pursued further.

Stop Consolidation Outcome

SBL used Troy and Athens High Schools as a sample for node-based stop consolidation. Only stops with assigned students were counted in the totals, excluding any start, way, or end points without students assigned. The next chart illustrates the number of students assigned per route, total stops before modification, active stops removed for each run, and remaining stops on routes compared to the baseline.

Route Set	Route	Assigned Students	Total Stops	Stops Removed	Stops Remain	Notes
AHS AM	01 AHS	62	5	-2	3	
AHS AM	03 AHS	104	10	-2	8	
AHS AM	11 AHS	58	6	-1	5	
AHS AM	12 AHS	78	7	-1	6	
AHS AM	17 AHS	65	11	-4	7	
AHS AM	22 AHS	99	9	-3	6	
AHS AM	25 AHS	72	10	-4	6	
AHS AM	26 AHS	59	7	-1	6	
AHS AM	49 AHS	75	5	0	5	
THS AM	04 THS	65	6	-3	3	
THS AM	05 THS	72	7	-2	5	Order changed
THS AM	07 THS	43	10	-1	9	
THS AM	08 THS	55	5	-1	4	Two stops combined @ MOR
THS AM	09 THS	79	5	-1	4	
THS AM	10 THS	46	7	0	7	Hazard stops – cannot consolidate due to safety
THS AM	14 THS	91	6	0	6	Could consolidate 2, pass intersect / evens load
THS AM	16 THS	87	10	-3	7	
THS AM	19 THS	47	2	0	2	2 stops on route / MOR Overflow
THS AM	20 THS	88	7	-3	4	
THS AM	23 THS	64	13	-1	12	
THS AM	24 THS	73	8	-2	6	
THS AM	28 THS	94	9	-4	5	
THS AM	31 THS	57	5	0	5	
THS AM	32 THS	111	11	-2	9	
THS AM	34 THS	60	6	-1	5	
THS AM	35 THS	91	10	-4	6	
THS AM	46 THS	72	7	-1	6	
THS AM	52 THS	82	9	-1	8	
THS AM	53 THS	68	8	-2	6	

Task 4 Summary:

- Shows 50 stops being removed from Athens and Troy High School routes
- Savings would be in the form of mileage/fuel reductions
- Labor time would have minimal reductions due to driver guarantees
- Route time savings would be in the form of deadhead time between runs

3.0 Summary

Baseline Summary

- 32 General Education Vehicles
- Daily labor hours of 153
- Daily mileage of 1161

Task 1 – Scenario A Summary:

- 39 vehicles are required for Gen Ed routing utilizing the 5-tier system (7 Additional Vehicles)
- Daily labor hours of 197
- Daily mileage of 1161

Task 2 – Scenario B Summary:

- 43 vehicles are required for Gen Ed routing utilizing the 5-tier system (11 Additional Vehicles)
- Daily labor hours of 215
- Daily mileage of 1161

Task 3 – Scenario C Summary:

- 56 vehicles are required for Gen Ed routing utilizing the 2-tier hybrid system (24 Additional Vehicles)
- Daily labor hours of 181
- Daily mileage of 1387

Task 4 – Scenario D Summary:

- Estimated 37-39 vehicles are required for Gen Ed routing utilizing the 5-tier system (5-7 Additional Vehicles)
- Estimated daily labor hours of 197
- Estimated daily mileage of 1161

4.0 Conclusion

The analysis presented in this report outlines the operational impacts, vehicle requirements, and labor considerations associated with each proposed scenario. While all scenarios explore later start times for high school students, each carries distinct trade-offs in terms of fleet utilization, driver hours, and overall efficiency.

School Bus Logistics is committed to supporting Troy School District in its efforts to balance student well-being, operational feasibility, and community needs while focusing on safety. We appreciate the opportunity to assist with this bell time study and look forward to any continued collaboration. Please do not hesitate to reach out with any follow-up questions or if further analysis is needed.